

# Hardaat Singh Baath

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## EDUCATION

### B.E. in Computer Science

Birla Institute of Technology and Science, Pilani ✉

CGPA: 7.55

Nov 2021 – Jul 2025

Sancoale, Goa, India

## EXPERIENCE

### Machine Learning Engineer

Nurix.AI ✉

Jan 2025 – present

Bangalore, India

- Developed an **Emotion Recognition Model** in **PyTorch** using **IEMOCAP**, **RAVDESS**, and **CREMA-D** datasets, achieving an **87% accuracy** in emotion classification.
- Implemented a **Multimodal** and **Multilingual Retrieval-Augmented Generation (RAG) pipeline**, optimizing inference speed and achieving **p95 latency of 2.93 seconds** with **96% accuracy**
- Contributed to the development of a **market research platform**, implementing **LLM driven** news analysis and action item generation pipelines

### AI Engineer

DG Takano ✉

May 2024 – Aug 2024

Tokyo, Japan

- Developed an automated testing system with **ESP32** for **Cloud Data logging**, improving accuracy and reducing manual effort.
- Integrated **Arduino Cloud** and **Slack Bots** for real-time alerts, enabling remote monitoring of the setup.
- Developed an **Automated G-code generator** that accelerated prototyping by learning from previous data, significantly improving the production of G-Code

### Research Intern

CSIR - CEERI, Pilani ✉

Jun 2023 – Aug 2023

Pilani, India

- Advisor: Dr. Dhiraj Sangwan ✉
- Developed a **DL-based pipeline** for damage generation, segmentation and restoration of Rajasthani Wall Murals.
- Used models like **StyleGAN2-ada-PyTorch** for damage generation, **UNet++** and **DeepLabV3+** for damage segmentation and **AOT-GAN** and **Partial Convolutional Network** for image inpainting.

## PUBLICATIONS

### Damage Segmentation and Restoration of Ancient Wall Paintings for Preserving Cultural Heritage ✉

Nov 2023

International Conference on Computer Vision and Image Processing, 2023

**Hardaat Singh Baath**, Soham Shinde, Jinam Keniya, Priyanshu Ranjan Mishra, Anil Saini, Dhiraj Sangwan

## PROJECTS

### Neuro Chaos Learning with Chaotic Maps

Jun 2024 – Nov 2024

Supervised Project

- Advisor: Dr. Harikrishnan NB ✉
- Explored the effect of various **chaotic maps** like Skew-tent, Gauss, Circle Maps etc for feature extraction for ML Applications.

- Analysed the effect of these chaotic features on the **speed and accuracy** of various ML models on standard datasets like iris, mnist, etc.
- Worked on designing a novel architecture that **leverages chaotic features** for enhanced performance and resilience to noise in dataset.

## Transformers for Stock Market Prediction

Feb 2024 – Nov 2024

### Supervised Project

- Advisor: Dr. J K Sahoo 
- Explored and implemented an **Transformers-based** time series algorithm for Stock Market Prediction using features like **Signal line, MacD line, EMA** etc with **CNN-based** feature extraction.
- Implemented a trading algorithm with transaction costs and taxation for realistic simulation of stock markets using **NIFTY 50** and **S&F 500** market datasets.

## Project Kratos, A Mars Rover

May 2023 – May 2024

### Club Project

- Worked on a **P-controlled visual servo** algorithm to follow Arrow and ArUco tags detected using **YOLOv3, YOLOv5, OpenCV and Python**.
- Implemented **RTK-GNSS**-based GPS coordinates system and **PID-based** algorithm to navigate to the specified points. Achieved accuracy of 10 cm.
- Implemented **Elevation Terrain Mapping** algorithm using Point Cloud information received from Zed2i stereo camera.
- Implementing **global and local path planning** algorithms for autonomous navigation in unknown environment Jetson Xavier AGX.

## Distributed Deployment Platform

Jan 2024 – May 2024

### Course Project

- Instructor: Dr Arnab K. Paul 
- Implemented a distributed deployment platform that deploys Github repositories on web using **AWS S3** and **ECS services**.
- Implemented backend with **API server** and **Reverse Proxy** using **NodeJS** for automated deployment and real time logs.
- Logs saved on **PostgreSQL Database**, using **Kafka Server** as an intermediate service.
- Added support for different types of languages like **ReactJS, NodeJS, HTML** etc to enable language based deployment.

## SKILLS

### Programming Languages

C, C++, Python, Java, NodeJS, JavaScript, MATLAB

### Libraries

Numpy, Matplotlib, Pandas, Seaborn, OpenCV, Scikit-learn

### Framework

ROS, PyTorch, Tensorflow, Keras

### Tools

Gazebo, Docker, AWS, Git and GitHub, Google Sheets, Arduino, Hugging Face

### Soft Skills

Public Speaking, Team Management, Creative Writing, Canva

### Languages

English, Hindi, Punjabi

## RELEVANT COURSES

### University Courses

- Computer Programming
- Data Structures and Algorithms
- Database Management Systems
- Operating Systems
- Computer Networks
- Computer Architecture
- Probability and Statistics
- Linear Algebra
- Differential Equations

- Multivariate Calculus
- Machine Learning
- Brain Inspired Deep Learning

### Online Courses

- IBM AI Engineering Specialization [↗](#)
- CS229 Stanford Machine learning [↗](#)
- CS231n Stanford Computer Vision [↗](#)
- AMRx: Autonomous Mobile Robotics, ETH Zurich [↗](#)

## AWARDS

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### National Qualifiers

Sep 2024

*NXP AIM 2024, Noida*

- Among **top 15 teams** out of 600+ teams that took part in NXP AIM 2024, held in Noida.
- Had to make an autonomous RC car with lane following, sign detection and obstacle avoidance.

### Stage 2 Qualifiers

Oct 2024

*LAM Research India 2024*

- Among **top 100** teams in India, out of 1000+ teams.
- Had to make a Peltier based cooling system with the ability to monitor and control the temperature.

### Excellence Award (Overall) [↗](#)

Jan 2023

*Internation Rover Competition, Bangalore*

- Won **Best Overall Rover** award and stood 2nd in the Autonomous Challenge and 4th overall.
- Had to make an Autonomous all terrain Rover with the ability to detect and follow arrows, while avoiding obstacles.

## MENTORSHIP AND VOLUNTEERING

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### PMP Mentor [↗](#)

Oct 2022 – Jul 2023

*Peer Mentorship Program, BITS Goa*

Sancoale, Goa, India

- Provided support to university freshmen, helping them navigate their first year through personalized counseling and guidance.
- Assisted students in both academic and non-academic areas, including study techniques, time management, and stress reduction strategies, so that they can get the best out of their first year.

### Subsystem Lead [↗](#)

May 2023 – May 2024

*Autonomous Subsystem, Project Kratos*

Sancoale, Goa, India

- Led a team of over 10 members in the Autonomous System, contributing to a student-run multidisciplinary project.
- Directed and managed various Research and Development projects, focusing on enhancing the current autonomous system.
- Collaborated with other subsystems to ensure seamless integration and functionality of the overall project.

### Project Mentor [↗](#)

Jun 2023 – Sep 2023

*Quark Summer Technical Projects, BITS Goa*

Sancoale, Goa, India

- Mentored a summer course on ROS (Robot Operating System), Robot Automation, and Computer Vision for university freshmen.
- Conducted instructional sessions and hands-on tutorials, providing students with practical experience and in-depth knowledge of the subjects.
- Evaluated weekly assignments, offering constructive feedback to help students improve their understanding and skills.

**Course Representative***Brain Inspired Deep Learning*

Jan 2024 – Jun 2024  
Sancoale, Goa, India

- Represented the students in the Brain Inspired Deep Learning course, acting as a liaison between students and faculty.
- Addressed student concerns and feedback, ensuring a positive and productive learning environment.
- Assisted in the implementation and coordination of course material and additional resources.

**Volunteer***Ironman 70.3*

Oct 2024  
Panjim, Goa, India

- Assisted in managing traffic to ensure the safety and smooth flow of the event.
- Provided support to cyclists, offering guidance and assistance as needed.
- Maintained clear and safe race tracks, ensuring no obstructions for participants.
- Collaborated with event organizers and other volunteers to ensure the event ran smoothly.
- Demonstrated strong organizational and communication skills in a high-pressure environment.